

```
. /* Dummy variable discussion
> Data file wagel.dta
> You are strongly encouraged to replicate the results in this handout
> */
```

```
. * read the data
. use wagel, clear
```

```
. sum
```

Variable	Obs	Mean	Std. dev.	Min	Max
wage	526	5.896103	3.693086	.53	24.98
educ	526	12.56274	2.769022	0	18
exper	526	17.01711	13.57216	1	51
tenure	526	5.104563	7.224462	0	44
nonwhite	526	.1026616	.3038053	0	1
female	526	.4790875	.500038	0	1
married	526	.608365	.4885804	0	1
numdep	526	1.043726	1.261891	0	6
smsa	526	.7224335	.4482246	0	1
northcen	526	.2509506	.4339728	0	1
south	526	.3555133	.4791242	0	1
west	526	.1692015	.3752867	0	1
construc	526	.0456274	.2088743	0	1
ndurman	526	.1140684	.318197	0	1
trcommpu	526	.0437262	.20468	0	1
trade	526	.2870722	.4528262	0	1
services	526	.1007605	.3012978	0	1
profserv	526	.2585551	.4382574	0	1
profocc	526	.3669202	.4824233	0	1
clerocc	526	.1673004	.3735991	0	1
servocc	526	.1406844	.3480267	0	1
lwage	526	1.623268	.5315382	-.6348783	3.218076
expersq	526	473.4354	616.0448	1	2601
tenursq	526	78.15019	199.4347	0	1936

```
. tab female
```

female	Freq.	Percent	Cum.
0	274	52.09	52.09
1	252	47.91	100.00
Total	526	100.00	

```
. * (1) To include a categorical variable, put an i. in front of the varname.
. * Stata calls this a factor variable.
. * Intercept dummy
. set showbaselevels off
```

```
. reg lwage educ exper tenure i.female
```

Source	SS	df	MS	Number of obs	=	526
				F(4, 521)	=	84.07
Model	58.1853046	4	14.5463261	Prob > F	=	0.0000
Residual	90.1444572	521	.173021991	R-squared	=	0.3923
				Adj R-squared	=	0.3876
Total	148.329762	525	.28253288	Root MSE	=	.41596

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0874623	.0069389	12.60	0.000	.0738307	.101094
exper	.0046294	.0016271	2.85	0.005	.0014328	.007826
tenure	.017367	.0029762	5.84	0.000	.0115201	.0232138
1.female	-.3011459	.0372456	-8.09	0.000	-.3743158	-.2279759
_cons	.5013479	.1019024	4.92	0.000	.3011579	.701538

```
. * (2) Interactions involving dummies
. * (a) interactions between dummies
. reg lwage educ exper tenure female married
```

Source	SS	df	MS	Number of obs	=	526
				F(5, 520)	=	70.38
Model	59.8675439	5	11.9735088	Prob > F	=	0.0000
Residual	88.4622178	520	.17011965	R-squared	=	0.4036
				Adj R-squared	=	0.3979
Total	148.329762	525	.28253288	Root MSE	=	.41246

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0839051	.0069728	12.03	0.000	.0702068	.0976035
exper	.003134	.0016821	1.86	0.063	-.0001705	.0064385
tenure	.016867	.0029554	5.71	0.000	.011061	.0226731
female	-.2855296	.0372643	-7.66	0.000	-.3587366	-.2123226
married	.1257391	.0399856	3.14	0.002	.0471859	.2042923
_cons	.4900581	.1011079	4.85	0.000	.291428	.6886882

```
. reg lwage educ exper tenure i.female i.married
```

Source	SS	df	MS	Number of obs	=	526
				F(5, 520)	=	70.38
Model	59.8675439	5	11.9735088	Prob > F	=	0.0000
Residual	88.4622178	520	.17011965	R-squared	=	0.4036
				Adj R-squared	=	0.3979
Total	148.329762	525	.28253288	Root MSE	=	.41246

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0839051	.0069728	12.03	0.000	.0702068	.0976035
exper	.003134	.0016821	1.86	0.063	-.0001705	.0064385
tenure	.016867	.0029554	5.71	0.000	.011061	.0226731

```

1.female | -.2855296 .0372643 -7.66 0.000 -.3587366 -.2123226
1.married | .1257391 .0399856 3.14 0.002 .0471859 .2042923
_cons | .4900581 .1011079 4.85 0.000 .291428 .6886882

```

```

. *           Now we will turn on the baselevels to see the "same" tables
. set showbaselevels on

```

```

. reg lwage educ exper tenure i.female i.married

```

```

Source |         SS          df           MS      Number of obs   =        526
-----+-----
Model | 59.8675439            5    11.9735088      F(5, 520)         =       70.38
Residual | 88.4622178          520     .17011965      Prob > F           =        0.0000
-----+-----
Total | 148.329762          525     .28253288      R-squared          =        0.4036
                                           Adj R-squared       =        0.3979
                                           Root MSE           =        .41246

```

```

lwage | Coefficient   Std. err.      t    P>|t|     [95% conf. interval]
-----+-----
educ | .0839051     .0069728     12.03  0.000     .0702068     .0976035
exper | .003134      .0016821      1.86  0.063    -.0001705     .0064385
tenure | .016867      .0029554      5.71  0.000     .011061     .0226731
female |
0 | 0 (base)
1 | -.2855296    .0372643     -7.66  0.000    -.3587366    -.2123226
married |
0 | 0 (base)
1 | .1257391     .0399856      3.14  0.002     .0471859     .2042923
_cons | .4900581     .1011079      4.85  0.000     .291428     .6886882

```

```

. *           Once you are comfortable with the interpretations, you can turn
baselevels off

```

```

. *           You can use different baselevels
. reg lwage educ exper tenure b1.female i.married

```

```

Source |         SS          df           MS      Number of obs   =        526
-----+-----
Model | 59.8675439            5    11.9735088      F(5, 520)         =       70.38
Residual | 88.4622178          520     .17011965      Prob > F           =        0.0000
-----+-----
Total | 148.329762          525     .28253288      R-squared          =        0.4036
                                           Adj R-squared       =        0.3979
                                           Root MSE           =        .41246

```

```

lwage | Coefficient   Std. err.      t    P>|t|     [95% conf. interval]
-----+-----
educ | .0839051     .0069728     12.03  0.000     .0702068     .0976035
exper | .003134      .0016821      1.86  0.063    -.0001705     .0064385
tenure | .016867      .0029554      5.71  0.000     .011061     .0226731
female |
0 | .2855296     .0372643      7.66  0.000     .2123226     .3587366
1 | 0 (base)

```

married							
0		0 (base)					
1	.1257391	.0399856	3.14	0.002	.0471859	.2042923	
_cons	.2045285	.0980315	2.09	0.037	.011942	.397115	

```

. *      interactions: ## or # connect the two categorical variables.
. *      ## include both main effects and interaction
. *      # include only interaction
. *      ## and # assume categorical variables, so no i. is needed.
. reg lwage educ exper tenure female##married

```

Source	SS	df	MS	Number of obs	=	526
				F(6, 519)	=	63.63
Model	62.8649837	6	10.4774973	Prob > F	=	0.0000
Residual	85.464778	519	.164672019	R-squared	=	0.4238
				Adj R-squared	=	0.4172
Total	148.329762	525	.28253288	Root MSE	=	.4058

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0835328	.0068608	12.18	0.000	.0700544	.0970112
exper	.0031638	.0016549	1.91	0.056	-.0000874	.006415
tenure	.015684	.0029209	5.37	0.000	.0099457	.0214223
female						
0	0 (base)					
1	-.0967496	.0574632	-1.68	0.093	-.2096386	.0161394
married						
0	0 (base)					
1	.2920907	.0553889	5.27	0.000	.1832767	.4009047
female#married						
1 1	-.3155762	.0739672	-4.27	0.000	-.460888	-.1702644
_cons	.3878165	.1023217	3.79	0.000	.1868009	.588832

```

. *      same as
. reg lwage educ exper tenure i.female i.married female#married

```

Source	SS	df	MS	Number of obs	=	526
				F(6, 519)	=	63.63
Model	62.8649837	6	10.4774973	Prob > F	=	0.0000
Residual	85.464778	519	.164672019	R-squared	=	0.4238
				Adj R-squared	=	0.4172
Total	148.329762	525	.28253288	Root MSE	=	.4058

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0835328	.0068608	12.18	0.000	.0700544	.0970112
exper	.0031638	.0016549	1.91	0.056	-.0000874	.006415

tenure		.015684	.0029209	5.37	0.000	.0099457	.0214223
female							
0		0	(base)				
1		-.0967496	.0574632	-1.68	0.093	-.2096386	.0161394
married							
0		0	(base)				
1		.2920907	.0553889	5.27	0.000	.1832767	.4009047
female#married							
1 1		-.3155762	.0739672	-4.27	0.000	-.460888	-.1702644
_cons		.3878165	.1023217	3.79	0.000	.1868009	.588832

. margins married##female

Predictive margins
Model VCE: OLS

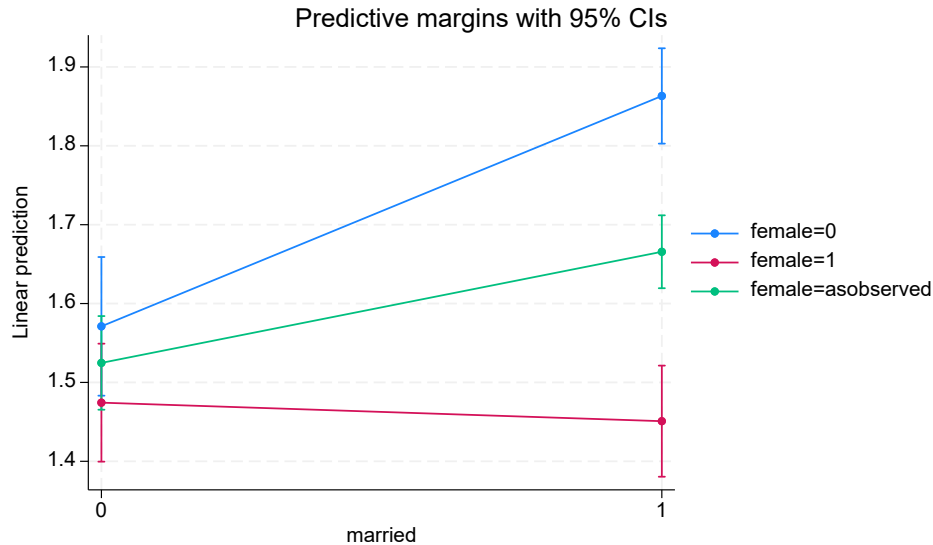
Number of obs = 526

Expression: Linear prediction, predict()

		Delta-method				[95% conf. interval]	
		Margin	std. err.	t	P> t		
married							
0		1.524765	.0302072	50.48	0.000	1.465421	1.584108
1		1.665667	.0235163	70.83	0.000	1.619468	1.711866
female							
0		1.748814	.0250706	69.76	0.000	1.699562	1.798066
1		1.460079	.0262541	55.61	0.000	1.408501	1.511656
married#female							
0 0		1.571116	.0447338	35.12	0.000	1.483235	1.658998
0 1		1.474367	.0380637	38.73	0.000	1.399589	1.549145
1 0		1.863207	.0307643	60.56	0.000	1.802769	1.923645
1 1		1.450881	.0358542	40.47	0.000	1.380444	1.521318

. marginsplot

Variables that uniquely identify margins: married female



```
. * (b) allowing different slopes - interactions with continuous variables
. reg lwage educ exper tenure i.married i.female
```

Source	SS	df	MS	Number of obs	=	526
Model	59.8675439	5	11.9735088	F(5, 520)	=	70.38
Residual	88.4622178	520	.17011965	Prob > F	=	0.0000
Total	148.329762	525	.28253288	R-squared	=	0.4036
				Adj R-squared	=	0.3979
				Root MSE	=	.41246

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0839051	.0069728	12.03	0.000	.0702068	.0976035
exper	.003134	.0016821	1.86	0.063	-.0001705	.0064385
tenure	.016867	.0029554	5.71	0.000	.011061	.0226731
married						
0	0	(base)				
1	.1257391	.0399856	3.14	0.002	.0471859	.2042923
female						
0	0	(base)				
1	-.2855296	.0372643	-7.66	0.000	-.3587366	-.2123226
_cons	.4900581	.1011079	4.85	0.000	.291428	.6886882

```
. * Now add female*educ
. reg lwage educ exper tenure i.married i.female female#c.educ
```

Source	SS	df	MS	Number of obs	=	526
Model	59.9126015	6	9.98543358	F(6, 519)	=	58.61
Residual	88.4171603	519	.170360617	Prob > F	=	0.0000
				R-squared	=	0.4039
				Adj R-squared	=	0.3970

Total | 148.329762 525 .28253288 Root MSE = .41275

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0865956	.0087211	9.93	0.000	.0694626	.1037286
exper	.0031475	.0016835	1.87	0.062	-.0001598	.0064547
tenure	.0169334	.0029603	5.72	0.000	.0111177	.0227491
married						
0	0	(base)				
1	.1255791	.0400151	3.14	0.002	.0469676	.2041907
female						
0	0	(base)				
1	-.1988898	.1725459	-1.15	0.250	-.5378642	.1400845
female#c.educ						
1	-.0069165	.0134489	-0.51	0.607	-.0333376	.0195045
_cons	.4550952	.121898	3.73	0.000	.2156211	.6945694

. margins female, dydx(educ)

Average marginal effects
Model VCE: OLS

Number of obs = 526

Expression: Linear prediction, predict()
dy/dx wrt: educ

		Delta-method				[95% conf. interval]	
		dy/dx	std. err.	t	P> t		
educ							
female							
0		.0865956	.0087211	9.93	0.000	.0694626	.1037286
1		.0796791	.0107804	7.39	0.000	.0585006	.1008576

. margins female, at(educ=(0(1)18))

Predictive margins
Model VCE: OLS

Number of obs = 526

Expression: Linear prediction, predict()

1._at: educ = 0
2._at: educ = 1
3._at: educ = 2
4._at: educ = 3
5._at: educ = 4
6._at: educ = 5
7._at: educ = 6
8._at: educ = 7
9._at: educ = 8
10._at: educ = 9

```

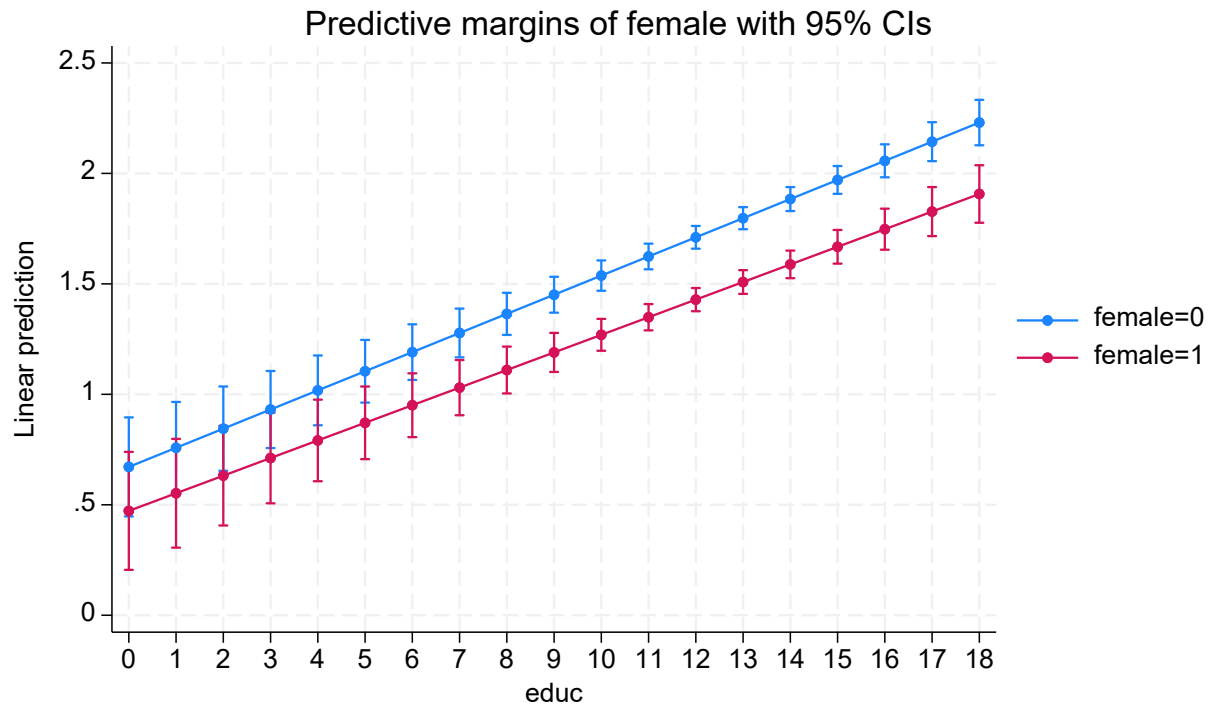
11._at: educ = 10
12._at: educ = 11
13._at: educ = 12
14._at: educ = 13
15._at: educ = 14
16._at: educ = 15
17._at: educ = 16
18._at: educ = 17
19._at: educ = 18

```

		Delta-method					
		Margin	std. err.	t	P> t	[95% conf. interval]	
_at#female							
1 0		.6714918	.1140486	5.89	0.000	.4474382	.8955455
1 1		.472602	.135904	3.48	0.001	.2056124	.7395916
2 0		.7580874	.1055633	7.18	0.000	.5507035	.9654713
2 1		.5522811	.1253475	4.41	0.000	.3060302	.7985319
3 0		.844683	.0971198	8.70	0.000	.6538867	1.035479
3 1		.6319601	.1148326	5.50	0.000	.4063663	.857554
4 0		.9312786	.08873	10.50	0.000	.7569644	1.105593
4 1		.7116392	.1043719	6.82	0.000	.5065959	.9166824
5 0		1.017874	.0804108	12.66	0.000	.8599036	1.175845
5 1		.7913182	.0939834	8.42	0.000	.6066836	.9759529
6 0		1.10447	.0721864	15.30	0.000	.9626562	1.246283
6 1		.8709973	.0836941	10.41	0.000	.7065765	1.035418
7 0		1.191065	.0640936	18.58	0.000	1.065151	1.31698
7 1		.9506764	.0735456	12.93	0.000	.8061927	1.09516
8 0		1.277661	.056189	22.74	0.000	1.167275	1.388047
8 1		1.030355	.0636054	16.20	0.000	.9053997	1.155311
9 0		1.364256	.0485648	28.09	0.000	1.268849	1.459664
9 1		1.1110035	.0539886	20.56	0.000	1.003971	1.216098
10 0		1.450852	.0413761	35.06	0.000	1.369567	1.532137
10 1		1.189714	.0449035	26.49	0.000	1.101499	1.277928
11 0		1.537448	.0348934	44.06	0.000	1.468898	1.605997
11 1		1.269393	.0367465	34.54	0.000	1.197202	1.341583
12 0		1.624043	.0295842	54.90	0.000	1.565924	1.682163
12 1		1.349072	.0302775	44.56	0.000	1.28959	1.408553
13 0		1.710639	.0261729	65.36	0.000	1.659221	1.762057
13 1		1.428751	.0267502	53.41	0.000	1.376199	1.481303
14 0		1.797234	.0254348	70.66	0.000	1.747267	1.847202
14 1		1.50843	.0273287	55.20	0.000	1.454741	1.562118
15 0		1.88383	.0275853	68.29	0.000	1.829637	1.938023
15 1		1.588109	.0317894	49.96	0.000	1.525657	1.650561
16 0		1.970426	.0320482	61.48	0.000	1.907465	2.033386
16 1		1.667788	.0388164	42.97	0.000	1.591531	1.744045
17 0		2.057021	.0380176	54.11	0.000	1.982334	2.131708
17 1		1.747467	.0472789	36.96	0.000	1.654586	1.840349
18 0		2.143617	.0448966	47.75	0.000	2.055415	2.231818
18 1		1.827146	.0565359	32.32	0.000	1.716079	1.938213
19 0		2.230212	.0523277	42.62	0.000	2.127412	2.333012
19 1		1.906825	.0662551	28.78	0.000	1.776664	2.036986

```
. marginsplot
```


Variables that uniquely identify margins: educ female



```
.
. * Notice that in the regression fem*edu is not significant.
. * So you can conclude that the slope for educ (return to education)
. * is the same for men and women. The two slopes are almost identical
. * in the graph.
.
. * Question: What if you want to test the poolability of men and women?
. *           In other words, you want to test the hypothesis that all
. *           the coefficients are the same between men and women.
.
. * (3) Interactions involving continous variables or polynomials
. reg lwage c.educ##c.exper tenure i.married i.female
```

Source	SS	df	MS	Number of obs	=	526
Model	59.8675489	6	9.97792481	F(6, 519)	=	58.54
Residual	88.4622129	519	.170447424	Prob > F	=	0.0000
Total	148.329762	525	.28253288	R-squared	=	0.4036
				Adj R-squared	=	0.3967
				Root MSE	=	.41285

lwage	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
educ	.0838553	.0115638	7.25	0.000	.0611377	.106573
exper	.0031058	.0054767	0.57	0.571	-.0076534	.013865
c.educ#c.exper	2.41e-06	.0004454	0.01	0.996	-.0008726	.0008775
tenure	.016866	.0029652	5.69	0.000	.0110408	.0226911

married							
0		0	(base)				
1		.1257404	.0400249	3.14	0.002	.0471097	.2043712
female							
0		0	(base)				
1		-.2855517	.0375237	-7.61	0.000	-.3592687	-.2118347
_cons		.4906912	.1547978	3.17	0.002	.186584	.7947984

```
. reg lwage c.educ##c.educ exper tenure i.married i.female
```

Source		SS	df	MS	Number of obs	=	526
					F(6, 519)	=	61.61
Model		61.7019014	6	10.2836502	Prob > F	=	0.0000
Residual		86.6278604	519	.166913026	R-squared	=	0.4160
					Adj R-squared	=	0.4092
Total		148.329762	525	.28253288	Root MSE	=	.40855

lwage		Coefficient	Std. err.	t	P> t	[95% conf. interval]
educ		-.01423	.0303975	-0.47	0.640	-.0739473 .0454873
c.educ#c.educ		.0041752	.0012595	3.32	0.001	.001701 .0066495
exper		.0027514	.0016701	1.65	0.100	-.0005297 .0060324
tenure		.0167322	.0029277	5.72	0.000	.0109805 .0224838
married						
0		0	(base)			
1		.128551	.039616	3.24	0.001	.0507234 .2063785
female						
0		0	(base)			
1		-.2706567	.037183	-7.28	0.000	-.3437044 -.1976089
_cons		1.030374	.1912971	5.39	0.000	.6545619 1.406186

```
. * The following command draws the relationship between lwage and educ
. margins, at(educ=(0(1)18))
```

Predictive margins
Model VCE: OLS

Number of obs = 526

Expression: Linear prediction, predict()

```
1._at: educ = 0
2._at: educ = 1
3._at: educ = 2
4._at: educ = 3
5._at: educ = 4
6._at: educ = 5
7._at: educ = 6
8._at: educ = 7
9._at: educ = 8
```

```

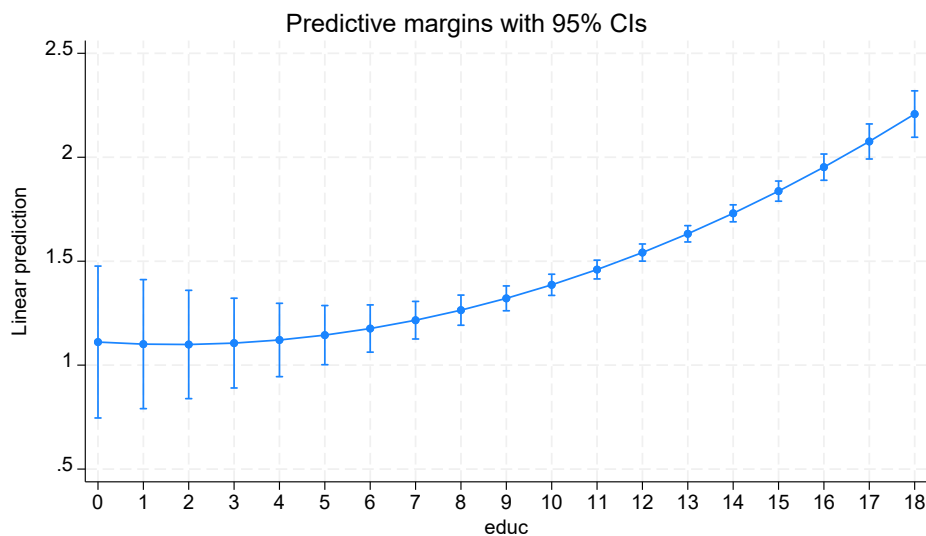
10._at: educ = 9
11._at: educ = 10
12._at: educ = 11
13._at: educ = 12
14._at: educ = 13
15._at: educ = 14
16._at: educ = 15
17._at: educ = 16
18._at: educ = 17
19._at: educ = 18

```

		Delta-method				
		Margin	std. err.	t	P> t	[95% conf. interval]
	_at					
	1	1.111143	.1859348	5.98	0.000	.7458653 1.47642
	2	1.101088	.1579784	6.97	0.000	.7907321 1.411443
	3	1.099383	.1326235	8.29	0.000	.8388385 1.359928
	4	1.106029	.1098952	10.06	0.000	.8901353 1.321923
	5	1.121026	.0898266	12.48	0.000	.9445575 1.297494
	6	1.144373	.0724597	15.79	0.000	1.002022 1.286723
	7	1.17607	.0578419	20.33	0.000	1.062437 1.289703
	8	1.216118	.0460066	26.43	0.000	1.125736 1.3065
	9	1.264516	.036922	34.25	0.000	1.191981 1.337051
	10	1.321264	.0303923	43.47	0.000	1.261557 1.380971
	11	1.386363	.025965	53.39	0.000	1.335354 1.437373
	12	1.459813	.0229991	63.47	0.000	1.41463 1.504995
	13	1.541612	.0209847	73.46	0.000	1.500387 1.582838
	14	1.631763	.0199694	81.71	0.000	1.592532 1.670993
	15	1.730263	.0208013	83.18	0.000	1.689398 1.771128
	16	1.837114	.0246708	74.47	0.000	1.788647 1.885581
	17	1.952316	.032113	60.80	0.000	1.889228 2.015403
	18	2.075868	.0429321	48.35	0.000	1.991526 2.16021
	19	2.20777	.0567904	38.88	0.000	2.096202 2.319337

```
. marginsplot
```

```
Variables that uniquely identify margins: educ
```



```
. * The following command draws the marginal effect of educ at each value
. margins, dydx(educ) at(educ=(0(1)18))
```

Average marginal effects
Model VCE: OLS

Number of obs = 526

Expression: Linear prediction, predict()

dy/dx wrt: educ

```
1._at: educ = 0
2._at: educ = 1
3._at: educ = 2
4._at: educ = 3
5._at: educ = 4
6._at: educ = 5
7._at: educ = 6
8._at: educ = 7
9._at: educ = 8
10._at: educ = 9
11._at: educ = 10
12._at: educ = 11
13._at: educ = 12
14._at: educ = 13
15._at: educ = 14
16._at: educ = 15
17._at: educ = 16
18._at: educ = 17
19._at: educ = 18
```

		Delta-method				
		dy/dx	std. err.	t	P> t	[95% conf. interval]
educ						
	_at					
	1	-.01423	.0303975	-0.47	0.640	-.0739473 .0454873
	2	-.0058796	.0279504	-0.21	0.833	-.0607893 .0490302
	3	.0024708	.0255172	0.10	0.923	-.0476588 .0526005
	4	.0108212	.0231024	0.47	0.640	-.0345644 .0562069

5		.0191717	.0207124	0.93	0.355	-.0215187	.059862
6		.0275221	.0183569	1.50	0.134	-.0085408	.0635849
7		.0358725	.0160511	2.23	0.026	.0043395	.0674055
8		.0442229	.0138198	3.20	0.001	.0170732	.0713726
9		.0525733	.011706	4.49	0.000	.0295764	.0755702
10		.0609237	.0097858	6.23	0.000	.0416992	.0801483
11		.0692742	.0081965	8.45	0.000	.0531718	.0853765
12		.0776246	.0071619	10.84	0.000	.0635546	.0916945
13		.085975	.006935	12.40	0.000	.0723509	.099599
14		.0943254	.0075884	12.43	0.000	.0794176	.1092332
15		.1026758	.0089311	11.50	0.000	.0851303	.1202213
16		.1110262	.0107067	10.37	0.000	.0899924	.1320601
17		.1193766	.0127355	9.37	0.000	.0943572	.1443961
18		.1277271	.0149145	8.56	0.000	.0984268	.1570273
19		.1360775	.0171867	7.92	0.000	.1023135	.1698415

```
. marginsplot
```

```
Variables that uniquely identify margins: educ
```

